

ARYA PRADEEP CHAVHAN

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EDUCATION

Sardar Patel Institute of Technology
BTech, Electronics and Telecommunication

2023 - Present
Overall CGPA: 9.16

Veermata Jijabai Technological Institute
Diploma, Electronics

2020 - 2023
Overall Percentage: 87.26

WORK EXPERIENCE

Smowcode
Embedded Software Developer Intern

June 2025 - July 2025

- Developed custom CAN (Controller Area Network) and OneWire protocol libraries for the ESP32 microcontroller from scratch, enabling reliable device communication
- Gained hands-on experience in low-level programming and protocol implementation, enhancing understanding of embedded communication systems
- Gained exposure to a fast-paced startup environment, fostering adaptability and initiative.

Brihanmumbai Electric Supply And Transport Undertaking
Student Trainee

June 2022 - August 2022

- Acquired comprehensive knowledge of SCADA systems, Telephone Exchange operations, and Wireless Communication technologies
- Gained exposure to real-world industrial environments, observing system functionality and maintenance processes

PROJECTS

FreeRTOS-Based Smart Home Security System with Live Dashboard

June 2025

- Built a FreeRTOS-based smart home system on ESP32 with sensors for temperature, humidity, pressure, motion, touch, and vibration.
- Enabled real-time monitoring and control via a web dashboard using task prioritization for sensor polling, interrupts, and server management.

Multimodal Emotion Detection from Image and Audio

April 2025

- Built a Python-based multimodal emotion detection system using DeepFace (facial) and logistic regression (speech) for emotion classification.

Mining Safety Wearables

May 2024

- Built a mining safety system with DHT11, MQ135, ultrasonic sensors, and a panic button for hazard detection.
- Used ESP32 with LoRa and GPS for real-time alerts and location tracking to improve emergency response.

Automatic Braking System

April 2023

- Developed an automatic braking system using IR and ultrasonic sensors for real-time hazard detection.
- Aimed to enhance vehicle safety by automatically triggering brakes to reduce collision risks.

TECHNICAL SKILLS

Languages	C, Micropython, Python, Assembly, Basic Java, Basic HTML, Basic CSS
Tools & Technologies	Keil uVision, Multisim, Proteus, ESP IDF, Arduino IDE, Thonny IDE, MATLAB, Simulink, FreeRTOS, IoT
Platforms	ESP32, ESP 8266, Arduino, Raspberry Pi, ARM LPC2148, PIC18F
Communication	UART, SPI, I2C, CAN, OneWire, ThingSpeak, Blynk, MQTT

EXTRACURRICULAR

IETE SPIT <i>Technical Team Member</i>	August 2024 - January 2025
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- Designed and programmed custom electronic circuits for competitions, ensuring precise functionality.
- Gained hands-on experience in PCB design basics and embedded system development.

Satkaam Foundation <i>Course Coordinator</i>	August 2024 - November 2024
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- Coordinated foundation-led courses in college, serving as the primary liaison between students and instructors.
- Facilitated smooth communication, scheduling, and updates between the Satkaam Foundation and participating students.

ACHIEVEMENTS

Ranked 2nd in Electronics Telecommunication Engineering (merit-wise) based on academic performance.

INTERESTS

Cooking, Reading, Listening to music, Gardening